

Maloney's Geomethry

An Axiomatic Formalization of Synthetic Geometry

$$\mathfrak{M} := \langle \mathbf{P}, \mathcal{B}, \cong, \mathcal{L}, \mathcal{A} \rangle$$

Daniel Maloney

July 19, 2026
v 0.0.1

Contents

0.1	Primitive Relations	iii
0.2	idk	iii
1	Primitive Notions	1
1.1	Plane	1
1.2	Points	1
1.3	Betweenness	2
1.3.1	Axioms of Betweenness	2
1.3.2	Immediate Consequences	3
1.4	Congruence	5
1.4.1	Axioms of Congruence	5
1.4.2	Congruence as an Equivalence Relation	6
1.4.3	Endpoint-Reversal Symmetries	7
1.5	Interaction of Betweenness and Congruence	8
2	Linear Notions	9
2.1	Segments	9
2.2	Rays	10
2.3	Lines	11
2.4	Midpoints and Bisectors	12
2.5	Parallelism	13
3	Partition of the Plane	16
3.1	Plane Separation Axiom	16
3.2	Same-Side Relation	16
3.3	Theorems	16
3.4	Half-Planes	17
4	Angles	19
4.1	Angles	19
4.2	Interior and Exterior of an Angle	19
4.3	Ray Betweenness	20
4.4	Congruence of Angles	21
4.5	Angle Construction	22
4.6	Angle Addition	23
5	Triangles	26
5.1	Defintion	26
6	Circles	27

7	Quadrilaterals	28
7.1	Definition	28
7.2	Diagonals	28
7.3	Trapezoid/Trapezium	28
7.3.1	Diagonals	29
7.3.2	Isosceles Trapezoid	29
8	Quadrilaterals	30
8.1	Definition	30
8.2	Diagonals	30
8.3	Trapezoid/Trapezium	30
8.3.1	Diagonal	30
8.3.2	Isosceles Trapezoid	30
8.3.3	Right Trapezoid	30
8.4	parallelograms	30
8.5	Special Parallelograms	31
8.5.1	Rectangles	31
8.5.2	Rhombus	31
8.5.3	Square	31
8.6	Parallelograms	31
8.7	Special Parallelograms	31
8.7.1	Rectangles	31
8.7.2	Rhombus	31
8.7.3	Square	31
8.8	Kites	31
8.8.1	Right Kite	33
8.9	Midpoint Theorems	33
8.10	Cyclic Quadrilaterals	33
8.11	Tangential Quadrilaterals	33
8.12	Bicentric Quadrilaterals	33
8.13	Other	33

Guide to Notation

Objects and Sets

$A, B, C, \dots, X, Y, Z$

Capital italic: **points**

P The set of all points

$\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \dots \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z}$ Capital x: geometric figures

$\overline{AB}$ **Line Segment**

$\overleftrightarrow{AB}$ **Line**

$\overrightarrow{AB}$ **Ray**

$\triangle ABC$ **Triangle**

$\angle ABC$ **Angle**

$\overline{AB} \parallel \overline{CD}$ **Parallel**

$\overline{AB} \perp \overline{CD}$ **Perpendicular**

0.1 Primitive Relations

$\mathcal{B}(A, B, C)$ Betweenness

$A = B$ Equivalence

$\overline{AB}, \overline{CD}$ Congruence

0.2 idk

$\mathcal{L}$ The **Language**

$\mathcal{A}$ The **Axiom Set**

Chapter 1

Primitive Notions

To build a deductive theory of geometry we must first fix the very language in which the theory will be expressed. Like any mathematical discipline, geometry begins with a collection of *primitive* concepts - terms that we do not define in terms of anything simpler but whose mutual relationships are governed by a system of axioms. From these axioms, we will derive theorems regarding the behavior and properties of these primitive notions. For our axiomatic system we adopt three primitive notions: one kind of object, and two relations among those objects. Together they form a single structure that captures the intuitive idea of a flat plane, and will later be used to prove represents a flat plane given our chosen axioms.

The first primitive notion is the **point**. Points are the individuals of our theory; they are the entities that occupy locations in the plane without themselves having parts or extension.

On this set we impose two primitive relations. The first is a *ternary* relation $\mathcal{B}$ (“betweenness”), and the second is a *quaternary* relation $\cong$ (“congruence”). These three ingredients - the point-set and the two relations - are the sole raw material from which the whole theory will be constructed.

The entire structure, consisting of the domain $\mathbf{P}$ equipped with the relations $\mathcal{B}$ and $\cong$, is what we call the **plane** and denote by $\mathcal{P}$.

1.1 Plane

Before dissecting the individual notions, it is useful to define the whole apparatus formally.

Definition 1 (Plane). *A plane is a structure*

$$\mathcal{P} := \langle \mathbf{P}, \mathcal{B}, \cong \rangle,$$

where $\mathbf{P}$ is an arbitrary set, $\mathcal{B}$ is a ternary relation on $\mathbf{P}$, and $\cong$ is a quaternary relation on $\mathbf{P}$.

Thus, saying “let $\mathcal{P}$ be a plane” amounts to specifying a set of points together with a particular interpretation of betweenness and congruence satisfying the axioms we shall gradually introduce. The definition deliberately uses the neutral term “a plane” rather than “the Euclidean plane” because, until we make further choices, the axioms we give will be true in a broad class of geometries.

1.2 Points

The elements of the domain $\mathbf{P}$ are termed *points*. We denote points by upper-case Latin letters: $A, B, C, \dots$. All points are gathered into a set $\mathbf{P}$, called the *point-set*, defined by

$$\mathbf{P} := \{ P \mid P \text{ is a point} \}.$$

Within the formal system, a point is an object whose only properties are those granted to it by the axioms governing $\mathcal{B}$ and $\cong$. There is no further internal structure: a point is defined not by decomposition into simpler parts, but solely by its membership in the set $\mathbf{P}$.

Axiom 1 (Existence of Distinct Points). $\exists A, B \in \mathbf{P}, A \neq B$.

This axiom guards against the trivial one-point model satisfying every later axiom.

Informal characterization. Although the formal theory treats points as primitives, it is helpful to keep in mind the intuitive picture that guides our choice of axioms. A point can be thought of as a unique, indivisible position in the plane. This corresponds to the classical Euclidean characterization of a point as “that which has no part.” Informally, a point is regarded as having no magnitude - no length, width, or depth - and as being zero-dimensional. However, this physical description is only a heuristic. Within the deductive framework, the formal behaviour of points is entirely determined by the relations in which they stand, and the intuitive characterization carries no logical weight in the proofs that follow. We draw on the geometric imagery only to motivate the axioms and to make the abstract relations vivid.

1.3 Betweenness

Betweenness is a ternary relation on the set of all points, denoted by $\mathcal{B} \subseteq \mathbf{P}^3$. Given points A , B , and C , we write $\mathcal{B}(A, B, C)$ and read it as “ B lies between A and C ”.

Unlike a set $\{A, B, C\}$, where the order of elements is immaterial, betweenness is defined over the cartesian product $\mathbf{P} \times \mathbf{P} \times \mathbf{P}$. Thus it depends on the order of the arguments. Formally,

$$\mathcal{B} \subseteq \mathbf{P}^3 := \{(A, B, C) \mid A, B, C \in \mathbf{P}\}.$$

The expression $\mathcal{B}(A, B, C)$ is simply the assertion that $(A, B, C) \in \mathcal{B}$.

In Euclidean geometry a point B is between A and C precisely when the three points lie on a line and B is met while traveling from A to C along that line. Our axioms capture this idea without invoking lines or order directly.

We now give five axioms governing the behavior of $\mathcal{B}$.

1.3.1 Axioms of Betweenness

Axiom 2 (Symmetry of Betweenness).

$$\forall A, B, C \in \mathbf{P}, \mathcal{B}(A, B, C) \implies \mathcal{B}(C, B, A).$$

Betweenness is symmetric with respect to the outer points: if B lies between A and C , then it also lies between C and A . This reflects the fact that the segment AC is the same as the segment CA . Left-to-right implication is used over a biconditional due to redundancy, as the right-to-left implication directly follows from this form of the axiom.

Axiom 3 (Order).

$$\forall A, B, X \in \mathbf{P}, \mathcal{B}(A, B, X) \implies (\neg \mathcal{B}(B, A, X) \wedge \neg \mathcal{B}(A, X, B)).$$

If B is between A and X , then A cannot be between B and X , nor can X be between A and B . In other words, a point can serve as a “middle” in only one way: the triple (A, B, X) determines a unique order along the line. Together with the Identity axiom, this implies that

$\mathcal{B}(A, A, A)$ is impossible (otherwise, taking $A = B = X$, we would have both $\mathcal{B}(A, A, A)$ and its negation). Thus the relation is *non-reflexive* in the strict sense: no point lies between itself and itself.

Axiom 4 (Identity of Betweenness).

$$\forall A, B \in \mathbf{P}, \mathcal{B}(A, B, A) \implies A = B.$$

This axiom says that if a point appears as the "middle" of a triple that begins and ends with the same point, then the two outer points must coincide. In particular, it forbids the possibility that a distinct point B lies between A and A ; such a configuration would contradict the intuition that betweenness requires a genuine interval. Notice that the converse, $A = B \implies \mathcal{B}(A, A, A)$, is *not* assumed; indeed the next axiom will exclude it.

Axiom 5 (Density).

$$\forall A, C \in \mathbf{P}, A \neq C \implies \exists B \in \mathbf{P}: \mathcal{B}(A, B, C).$$

Whenever two points are distinct, there exists a point strictly between them. This captures the idea that segments are dense—between any two points one can always find a third.

The plane must be infinite. Thus, we introduce an axiom that will later allow us to guarantee the existence of infinite points along a line.

Axiom 6 (Extensibility). *Given two points A and B , there exists a point C such that $\mathcal{B}(A, B, C)$*

$$\forall A, B \in \mathbf{P}, A \neq B \implies \exists C \in \mathbf{P}: \mathcal{B}(A, B, C).$$

Axiom 7 (Transitivity of Betweenness). $\mathcal{B}(A, B, C) \wedge \mathcal{B}(B, C, D) \wedge B \neq C \implies \mathcal{B}(A, B, D)$.

If B lies between A and C , and C lies between B and D (with $B \neq C$), then B lies between A and D . This axiom allows us to chain betweenness relations along a line.

These six axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

1.3.2 Immediate Consequences

As is, the extensibility axiom allows only for extensibility to the right. To prove this consisely for both directions we introduce a lemma demonstrating extensibility on the left side:

Theorem 1 (Left-Sided Extensibility). *For any two points A and B , there exists C such that $\mathcal{B}(C, B, A)$.*

Proof. From $A \neq B$, the extensibility axiom 6 implies $\mathcal{B}(A, B, C)$. We then apply the symmetry of betweenness 2 to obtain $\mathcal{B}(C, B, A)$, thus arriving at the desired result. $\square$

With this, we can now prove extensibility on both sides.

Theorem 2 (Dual-Sided Extensibility). *Given any two points B and C , there exists points A and D such that $[ABCD]$.*

Proof. Let $B, C \in \mathbf{P}$ be two points not equal to each other. From the extensibility axiom, there exists a point D such that $[BCD]$. Similarly, from left-sided extensibility, there exists a point A such that $[ABC]$. These results are sufficient to demonstrate that $[ABCD]$, thus proving the theorem. $\square$

Theorem 3. $\mathcal{B}(A, B, C) \iff \mathcal{B}(C, B, A)$ for all $A, B, C \in \mathbf{P}$.

Proof. The forward direction is the axiom; the reverse follows by applying the axiom with A and C exchanged. $\square$

Given two distinct points, there is always a point beyond the second, so that the second lies between the first and the new point. This guarantees that lines extend indefinitely.

These five axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

Theorem 4 (Irreflexivity of Betweenness). *No point lies between it and itself:*

$$\forall A \in \mathbf{P}, \neg \mathcal{B}(A, A, A).$$

Proof. Assume $\mathcal{B}(A, A, A)$. By the order axiom, this implies $\neg \mathcal{B}(A, A, A)$ and $\neg \mathcal{B}(A, A, A)$, which contradicts our assumption. Hence, $\neg \mathcal{B}(A, A, A)$. $\square$

Theorem 5 (Distinct Endpoints). *Betweenness forces distinct outer points:*

$$\mathcal{B}(A, B, C) \implies A \neq C.$$

Proof. Suppose $\mathcal{B}(A, B, C)$ and $A = C$. Then, we have $\mathcal{B}(A, B, A)$, which by the identity of betweenness implies $A = B$. Substituting back gives $\mathcal{B}(A, A, A)$, which contradicts the irreflexivity of betweenness. Therefore $A \neq C$. $\square$

Theorem 6 (Irreflexivity of Betweenness for Distinct Points).

$$A \neq B \implies \neg \mathcal{B}(A, A, B) \wedge \neg \mathcal{B}(A, B, B).$$

Proof. If $\mathcal{B}(A, A, B)$, then by the order of betweenness, $\square$

Theorem 7 (Strict Betweenness). *If $\mathcal{B}(A, B, C)$, then $A \neq B$ and $B \neq C$.*

Proof. Suppose $\mathcal{B}(A, B, C)$ and $A = B$. Then, $\mathcal{B}(A, A, C)$, which implies $\neg \mathcal{B}(A, A, C)$ by the order of betweenness. However, this is a contradiction. Thus, $A \neq B$. Similarly, if we assume $B = C$, then we have $\mathcal{B}(A, B, B) \implies \neg \mathcal{B}(A, B, B)$, leading to a contradiction. Thus, $B \neq C$. $\square$

Theorem 8 (Contrapositive Ordering). *For all $A, B, X \in \mathbf{P}$,*

$$\mathcal{B}(B, A, X) \implies \neg \mathcal{B}(A, B, X) \quad \text{and} \quad \mathcal{B}(A, X, B) \implies \neg \mathcal{B}(A, B, X).$$

Proof. The Order axiom gives the two implications

$$\mathcal{B}(A, B, X) \implies \neg \mathcal{B}(B, A, X) \quad \text{and} \quad \mathcal{B}(A, B, X) \implies \neg \mathcal{B}(A, X, B).$$

Taking the logical contrapositive of each implication (which is logically equivalent to the original), yields precisely the two stated results. $\square$

Theorem 9 (Symmetry Extended). *Given $\mathcal{B}(A, B, C)$ none of the following hold:
 $\mathcal{B}(B, A, C)$, $\mathcal{B}(A, C, B)$, $\mathcal{B}(B, C, A)$, $\mathcal{B}(C, A, B)$.*

These theorems complete our baseline analysis of how the betweenness relation $\mathcal{B}$ operates locally on individual triples of points. However, proving that the plane $\mathcal{P}$ is structurally infinite—both in its internal density and its outward extension—requires the formal definitions of segments and lines. We defer these proofs to chapter blank, where the infinite nature of the geometric line can be properly established.

1.4 Congruence

Congruence is a quaternary relation on the set of all points denoted by $\cong \subseteq \mathbf{P}^4$. Rather than mapping individual points to one another, this relation establishes a geometric equivalence between two pairs of points. We write $\cong (A, B, C, D)$, to assert that the segment from point A to point B is congruent to the segment from point C to point D . Formally:

$$\cong (A, B, C, D) \iff (A, B, C, D) \in \cong .$$

To reflect this conceptually, we bundle the point pairs (A, B) and (C, D) into the typographical units AB and CD and introduce the infix symbol $\cong$:

$$AB \cong CD \stackrel{\text{def}}{\iff} \cong (A, B, C, D)$$

If the relation holds for the pairs (A, B) and (C, D) , we write $(A, B) \cong (C, D)$, which is interpreted as the "distance" from A to B being equal to the distance from C to D . The notation for the points will be reused to define segments. This choice was intentional, as the later informal usage of congruence upon segments is identical in nature to what is presented below.

We now impose three axioms governing the behavior of $\cong$.

1.4.1 Axioms of Congruence

Axiom 8 (Reversal of Congruence).

$$\forall A, B \in \mathbf{P}, AB \cong BA.$$

Axiom 9 (Transitivity of Congruence).

$$\forall A, B, C, D, E, F \in \mathbf{P}, (AB \cong CD \wedge AB \cong EF) \implies CD \cong EF.$$

Transitivity (Axiom 9) is the formal way of saying that two segments congruent to the same segment are congruent to each other.

With only Reversal and Transitivity, however, the axioms do not distinguish between genuine segments (whose endpoints are distinct) and degenerate ones (whose endpoints coincide). Model in which every point pairs is congruent to one another would strictly satisfy the above. In this universe, distinct points exist (Axiom 1), but they are not separated. Such a “zero-distance” universe reduces the congruence relation to the universal relation and makes the geometry collapse: all points are at the same distance from one another, and no meaningful notion of length can be recovered.

To exclude this degenerate model we introduce the following.

Axiom 10 (Identity of Congruence).

$$\forall A, B, C \in \mathbf{P}, AB \cong CC \implies A = B.$$

In words: if a segment is congruent to a degenerate segment, then its endpoints must coincide. Equivalently, a non-degenerate segment ($A \neq B$) can never be congruent to a degenerate one (CC). Intuitively, this means the "distance" between distinct points is never zero, — a requirement that, in a metric geometry, would be imposed by the condition $d(A, B) = 0 \iff A = B$.

Note that the axiom does not guarantee the existence of distinct points — that guarantee was already given by the domain axiom Axiom 1. However, these axioms together guarantee that the plane contains at least two points *and* that distinct points are geometrically distinct.

Once distinct, genuinely separated points exist, ?? populates the segment between them with infinitely many intermediate points, ensuring that the line is not a discrete set of isolated locations.

1.4.2 Congruence as an Equivalence Relation

An equivalence relation is characterized by three properties: reflexivity, symmetry, and transitivity. The axioms adopted here for $\cong$ assert neither reflexivity nor the general symmetry and transitivity required of an equivalence relation directly; Transitivity, in particular, is stated in a shared-first-argument form rather than the familiar chain form. The remainder of this subsection shows that Reversal and Transitivity nonetheless force Reflexivity, Reflexivity together with Transitivity forces Symmetry, and Symmetry together with Transitivity forces the familiar chain form of transitivity, so nothing further needs to be assumed.

Theorem 10 (Reflexivity of Congruence).

$$\forall A, B \in \mathbf{P}, AB \cong AB.$$

Proof. From reversal (Axiom 8), we have $AB \cong BA$. Through replacement of variables we also have $BA \cong AB$. We now instantiate transitivity (Axiom 9) with $AB \cong BA$ used twice:

$$(AB \cong BA \text{ and } AB \cong BA) \implies BA \cong BA.$$

Next, we apply transitivity again as follows:

$$(BA \cong AB \text{ and } BA \cong AB) \implies AB \cong AB,$$

which completes the proof. □

With reflexivity established, the other expected properties follow rapidly.

Theorem 11 (Symmetry of Congruence). *If $AB \cong CD$, then $CD \cong AB$.*

Proof. We are given $AB \cong CD$. From the reflexivity axiom we also have $AB \cong AB$. Applying these two congruences to the transitivity axiom gives the antecedent

$$AB \cong CD \wedge AB \cong AB,$$

which implies $CD \cong AB$. □

Theorem 12 (Full-Transitivity of Congruence). *If $AB \cong CD$ and $CD \cong EF$, then $AB \cong EF$.*

Proof. By hypothesis, $AB \cong CD$. Applying Symmetry (Theorem 11) to this gives

$$CD \cong AB.$$

Together with the second hypothesis $CD \cong EF$, we obtain the conjunction

$$CD \cong AB \wedge CD \cong EF.$$

This is precisely an instance of the Transitivity axiom (Axiom 9), instantiated with CD in the shared first-argument position of both conjuncts. The axiom therefore yields

$$AB \cong EF,$$

as required. □

With these results, $\cong$ is established as an equivalence relation on ordered pairs of points: reflexive and symmetric by theorem, and transitive in the familiar chain sense by Full-Transitivity. Consequently, $\cong$ partitions the set of all ordered point pairs into disjoint equivalence classes, each consisting of all segments congruent to one another — the geometric counterpart of segments of equal length. This is the extent of what an equivalence relation alone provides: it identifies which segments share a length, but supplies no order between distinct classes and no arithmetic operation upon them. Such structure requires the interaction of $\cong$ with $\mathcal{B}$ and is developed in a later section.

1.4.3 Endpoint-Reversal Symmetries

The equivalence-relation properties established above govern the behavior of $\cong$ across distinct pairs of points; they say nothing about the effect of reordering the two endpoints *within* a single pair. This subsection establishes that $\cong$ is in fact invariant under such reordering, in every combination.

Theorem 13 (Left-Reversal of Congruence). *If $AB \cong CD$, then $BA \cong CD$.*

Proof. We are given $AB \cong CD$. From the reversal axiom, we also have $AB \cong BA$. Applying transitivity to the conjunction

$$AB \cong BA \wedge AB \cong CD$$

yields $BA \cong CD$. □

Theorem 14 (Right-Reversal of Congruence). *If $AB \cong CD$, then $AB \cong DC$.*

Proof. We are given $AB \cong CD$. From the reversal axiom applied to the pair (C, D) , we have $CD \cong DC$. By symmetry (Theorem 11) applied to the hypothesis, we also have $CD \cong AB$. Applying transitivity to the conjunction

$$CD \cong AB \wedge CD \cong DC,$$

yields $AB \cong DC$. □

Theorem 15 (Full-Reversal of Congruence). *If $AB \cong CD$, then $BA \cong DC$.*

Proof. We are given $AB \cong CD$. From the reversal axiom, we know that $AB \cong BA$. By Right-Reversal (Theorem 14) applied to the hypothesis, we also have $AB \cong DC$. Applying transitivity to

$$AB \cong BA \wedge AB \cong DC,$$

yields $BA \cong DC$. □

The previous three theorems can be packaged into a single convenient equivalence, establishing that segment congruence is entirely indifferent to the order of the endpoints of either segment.

Theorem 16 (Endpoint-Swap Invariance). *For any points A, B, C, D ,*

$$AB \cong CD \iff BA \cong DC \iff AB \cong DC \iff BA \cong CD.$$

Proof. The forward implications have all been proved:

- $AB \cong CD \Rightarrow BA \cong DC$ (Full-Reversal)
- $BA \cong DC \Rightarrow AB \cong DC$ (apply Left-Reversal to the pair (B, A))
- $AB \cong DC \Rightarrow BA \cong CD$ (Right-Reversal followed by Left-Reversal)
- $BA \cong CD \Rightarrow AB \cong CD$ (Left-Reversal on (B, A))

The reverse directions follow by the same theorems applied in the opposite order, completing the circle of equivalences. □

1.5 Interaction of Betweenness and Congruence

Axiom 11 (Segment Construction).

$$\forall Q, X, A, B \in \mathbf{P}, Q \neq X \implies \exists Y \in \mathbf{P}: \mathcal{B}(Q, X, Y) \wedge XY \cong AB.$$

Chapter 2

Linear Notions

Now that the domain and relations of our plane have been fleshed-out, we set off to define objects present on the plane. That is, sets of points that meeting a defined criteria. Objects like this are called *figures*, wherein a figure F on the plane is a subset of the point set: $F \subseteq \mathbf{P}$. Note that a figure is a proper subset of $\mathbf{P}$, as the set containing no points is simply nothing, and the set containing all points is the domain itself.

The most fundamental types of geometric figures are the segment, ray, and line - which we classify as *linear notions*. In this chaoter, we define all three linear notions, and detirmine their basic properties.

Theorem 17 (Infinite interior points). *Let $A, C \in \mathbf{P}$ with $A \neq C$. Then there exist infinitely many points strictly between A and C .*

Proof. wasd □

Theorem 18 (Infinite outward extension). *Let $A, B \in \mathbf{P}$ with $A \neq B$. Then there exist infinitely many points $C_1, C_2, C_3, \dots$ such that*

$$\mathcal{B}(A, B, C_1), \quad \mathcal{B}(A, C_1, C_2), \quad \mathcal{B}(A, C_2, C_3), \quad \dots$$

Proof. wasd □

Together, these two lemmas show that from any segment AB ($A \neq B$) we can generate infinitely many points in both directions along the line determined by A and B . Thus the plane $\mathcal{P}$ is necessarily infinite.

2.1 Segments

A *line segment* AB is the set containing A , B , and all points between A and B . The points A and B are called the *endpoints* of the segment. Symbolically, that is

Definition 2 (Segment).

$$\forall A, B \in \mathbf{P}, A \neq B \implies AB := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

By intuition, we can see that the segment AB is identical to the segment BA . Using properties of sets upon the endpoints amd the symmetry of betweeness 2 upon the point in between X we can prove this result.

Theorem 19 (Segment Symmetry). *For any two points A, B , the segment AB is equal and equivalent to the segment BA .*

$$\forall A, B \in \mathbf{P}, AB = BA.$$

Proof. By the definition of a line segment, we have:

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$$

and

$$BA = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we examine the sets of interior points. By the **Symmetry of Betweenness**, the relation $\mathcal{B}(A, X, B)$ holds if and only if $\mathcal{B}(B, X, A)$ holds. Formally, this implies:

$$\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

Since AB and BA are defined as the union of identical sets, they contain the same points. By the principle of extensionality in set theory, $AB = BA$. $\square$

2.2 Rays

We now consider the case where a line segment extends out in one direction. A ray pointing from point A to a distinct point B is a subset of $\mathbf{P}$ denoted as $\overrightarrow{AB}$. It is defined as the union of A , B , all points between A and B , and all points such that B is between A and that point, that is

Definition 3 (Ray).

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The point the ray begins at is called the idk. Our notation will allow for the arrow to be facing left or right. However, it is preferred to have the arrow pointing right.

Theorem 20 (Ray Asymmetry). *For any two distinct points $A, B \in \mathbf{P}$, the ray $\overrightarrow{AB}$ is not equal to the ray $\overrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \neq \overrightarrow{BA}.$$

Proof. We proceed by contradiction. Assume that the rays for some distinct points $A, B \in \mathbf{P}$, $\overrightarrow{AB} = \overrightarrow{BA}$.

By the definition of a ray, we have:

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\},$$

and

$$\overrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

By the **Outer Point Axiom**, there must exist a point Y such that $\mathcal{B}(A, B, Y)$. By the definition of $\overrightarrow{AB}$, it follows that $Y \in \overrightarrow{AB}$. Under our initial assumption that the rays, Y must be an element of $\overrightarrow{BA}$.

By the Distinctness of Betweenness, the relation $\mathcal{B}(A, B, Y)$ implies that $A \neq Y$ and $B \neq Y$. Therefore, $Y \notin \{A, B\}$.

Since $Y \in \overrightarrow{BA}$ but is not an endpoint, it must satisfy $\mathcal{B}(B, Y, A)$ or $\mathcal{B}(B, A, Y)$.

However, because we know that $\mathcal{B}(A, B, Y)$ is true, by Order of Betweenness we have $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(B, A, Y)$, and $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(A, Y, B)$, which by the Symmetry of Betweenness is logically equivalent to $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(B, Y, A)$.

Thus, both conditions for Y to be an element of $\overrightarrow{BA}$ are false. Therefore, the rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are not equal, thus proving the desired result. $\square$

Theorem 21 (Union of Opposite Rays). *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cup \overrightarrow{AB} = \overleftrightarrow{AB}.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The union of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{A, B\}$, and all points X such that $\mathcal{B}(X, A, B)$ or $\mathcal{B}(A, X, B)$, or $\mathcal{B}(A, B, X)$.

Note that this is the same as the line $\overleftrightarrow{AB}$, as it is defined as the set

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

This set is equal to the union described above. Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the line $\overleftrightarrow{AB}$. $\square$

Theorem 22 (Intersection of Opposite Rays). *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cap \overrightarrow{AB} = AB.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The intersection of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{A, B\}$ and all points X between A and B , that is $\mathcal{B}(A, X, B)$.

This intersection consists of the same points contained in the segment AB , as

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the segment AB . $\square$

2.3 Lines

The line $\overleftrightarrow{AB}$ is the subset of $\mathbf{P}$ consisting of the points A, B , all points between A and B , all points where A is between it and B , and all points where B is between it and A . That is,

Definition 4 (Line).

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overleftrightarrow{AB} := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

Theorem 23 (Line Symmetry). *For any two distinct points A and B , the line $\overleftrightarrow{AB}$ is equal to the segment $\overleftrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, \overleftrightarrow{AB} = \overleftrightarrow{BA}.$$

Proof. By the definition of a line, we have:

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}$$

and

$$\overleftrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A) \vee \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we demonstrate that the three cases of betweenness in $\overleftrightarrow{AB}$ are logically equivalent to those of $\overleftrightarrow{BA}$ using the by the **Symmetry of Betweenness**:

- $\{X \in \mathbf{P} \mid \mathcal{B}(X, A, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, A, X)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, B, X)\} = \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A)\}$

Since each condition for membership in $\overleftrightarrow{AB}$ is logically equivalent to a condition for membership in $\overleftrightarrow{BA}$, the sets contain the same points. Thus, by the principle of extensionality, $\overleftrightarrow{AB} = \overleftrightarrow{BA}$ □

2.4 Midpoints and Bisectors

Intuitively, the midpoint of a segment is its “center” - the point that splits the segment into two equal halves. In real life, you find a midpoint when you fold a piece of paper so that the ends meet; the crease marks the exact middle. But in our geometry, which is built only on points, betweenness, and congruence, we need to be precise: a point is the midpoint of segment AB exactly when it lies on the segment (so it is between A and B) and the two subsegments it creates are congruent.

However, we must ensure that such a point always exists and that there is only one. Geometry would be very strange if some segments could not be cut in half, or if a segment had two distinct “centers.” We introduce two axioms to guarantee this, but first present a formal definition of a midpoint.

Definition 5 (Midpoint). *For distinct points $A, B \in \mathbf{P}$, a point M is a midpoint of segment AB if*

$$\mathcal{B}(A, M, B) \quad \text{and} \quad AM \cong MB.$$

Now, the axioms:

Axiom 12 (Midpoint Existence). $\forall A, B \in \mathbf{P}, A \neq B \implies \exists M \in \mathbf{P} : (\mathcal{B}(A, M, B)) \wedge (AM \cong MB)$.

Axiom 13 (Midpoint Uniqueness). $\forall A, B \in \mathbf{P}, A \neq B \implies \forall M, N \in \mathbf{P} : (AM \cong MB \wedge \mathcal{B}(A, M, B) \wedge AN \cong NB \wedge \mathcal{B}(A, N, B)) \implies M = N$.

The first ensures that given two points A and B , there exists a point M which is the midpoint of the segment AB . However, this alone allows for geometries in which segments have multiple midpoints. Thus, we introduce an axiom ensuring the uniqueness of the midpoint stating that for segment AB , if points M and N both meet the criteria for being a midpoint, then it is necessary that $M = N$. These axioms therefore require existence of a unique midpoint given any line segment.

Now we ask: what happens if we drop the requirement that the point lies on the segment, but keep the requirement for congruence. That is, we determine all points X such that $AX \cong XB$. In general, a line containing the midpoint of a segment is called a bisector of that segment. However, this set of points gets its own name.

Definition 6 (Perpendicular Bisector). *For distinct points $A, B \in \mathbf{P}$, the perpendicular bisector of segment AB is the set*

$$\ell_{AB} := \{ X \in \mathbf{P} \mid AX \cong XB \}.$$

The name is justified by two facts (which we will prove later):

1. ℓ_{AB} is a line (i.e., there exist points P, Q such that $\ell_{AB} = \overleftrightarrow{PQ}$);
2. ℓ_{AB} is perpendicular to the line $\overleftrightarrow{AB}$ and passes through the unique midpoint M of AB .

Notice that the midpoint M itself belongs to ℓ_{AB} because $AM \cong MB$, and M is the only point of ℓ_{AB} that lies on the segment AB (the betweenness condition forces it to be between A and B). The rest of the perpendicular bisector lies off the segment, forming the two rays that are perpendicular to AB at M .

Remark. Defining the perpendicular bisector as the equidistant set $\{ X \mid AX \cong XB \}$ has the advantage of making no prior reference to perpendicularity or to the midpoint; both properties are then *derived* from the congruence and betweenness axioms. This is the standard treatment in absolute (neutral) geometry.

2.5 Parallelism

Having introduced the fundamental linear figures—segments, rays, and lines—we now turn our attention to a relation that captures one of the most essential geometric intuitions: the notion that two lines “point in the same direction” and therefore never meet. This relation is called *parallelism*. In a purely set-theoretic framework, where lines are subsets of the point set $\mathbf{P}$, the idea that two distinct lines have no point in common is naturally expressed as the emptiness of their intersection. There is, however, a subtlety concerning the limiting case in which the two lines are actually the same set: should a line be considered parallel to itself? Different axiomatic traditions answer this differently. In the ordinary language of school geometry one often says that parallel lines are distinct lines that do not intersect, thereby excluding the reflexive case. In a formal development, however, it is far more convenient to adopt the inclusive convention and allow a line to be parallel to itself. Doing so makes parallelism an equivalence relation—a structure that will prove invaluable later when we classify lines by their *direction*. The inclusive definition is the one we adopt here; it simplifies the statement of theorems.

Because parallelism is a relation that compares lines, we must first collect all lines into a single object that can serve as the domain of the relation. Our definition of a line $\overleftrightarrow{AB}$ yields a subset of $\mathbf{P}$ for every pair of distinct points A, B ; the collection of all such subsets is therefore a well-defined family of sets. We denote this family by $\mathbf{L}$ and make it the official set of lines of our plane.

Definition 7 (Set of all lines). *The set of all lines $\mathbf{L}$ is defined as*

$$\mathbf{L} := \{ l \subseteq \mathbf{P} \mid \exists A, B \in \mathbf{P}, A \neq B \wedge l = \overleftrightarrow{AB} \}.$$

Every element of $\mathbf{L}$ is a subset of $\mathbf{P}$ that arises from two distinct points by the construction of the line; conversely, every such construction yields a member of $\mathbf{L}$. This definition guarantees that we have a fixed, well-defined set to talk about, a necessary prerequisite for building a binary relation.

Now we are in a position to define parallelism itself. The definition is a direct set-theoretic translation of the geometric idea: two lines are parallel if they coincide or have no points in common.

Definition 8 (Parallel lines). *Let $l, m \in \mathbf{L}$ be lines. We say that l is parallel to m , and write $l \parallel m$, if and only if*

$$l = m \quad \text{or} \quad l \cap m = \emptyset.$$

Formally,

$$\forall l, m \in \mathbf{L}, \quad l \parallel m \iff (l = m) \vee (l \cap m = \emptyset).$$

The first disjunct, $l = m$, may at first seem surprising if one is accustomed to thinking of parallel lines as distinct. Its inclusion is deliberate. By allowing a line to be parallel to itself we immediately obtain reflexivity (every line is parallel to itself), and together with the natural symmetry and a suitable parallel postulate we shall obtain transitivity, turning $\parallel$ into an equivalence relation on $\mathbf{L}$. The equivalence classes of this relation are precisely the *directions* in the plane—a concept that lies at the heart of many later arguments. If one preferred the exclusive convention, one could define a separate relation “strictly parallel” and then prove that it differs from $\parallel$ only by missing the diagonal of $\mathbf{L} \times \mathbf{L}$; the two treatments are completely intertranslatable, and we find the inclusive one more elegant.

In the absence of any further axioms, this definition tells us very little about how many parallels exist through a point not on a given line. In fact, neutral (absolute) geometry guarantees the existence of *at least one* such parallel, but leaves the number undetermined. To obtain the familiar Euclidean picture, we shall later introduce an additional postulate—Playfair’s Axiom—that asserts the uniqueness of the parallel. Until then, the definition stands as a purely combinatorial condition on subsets of the plane.

Theorem 24 (Reflexivity of Parallelism). *For every line $l \in \mathbf{L}$, $l \parallel l$.*

Proof. Immediate from the definition: take $l = m$ in the disjunct $l = m$. □

Theorem 25 (Symmetry of Parallelism). *For all lines $l, m \in \mathbf{L}$,*

$$l \parallel m \implies m \parallel l.$$

Proof. Assume $l \parallel m$. By definition, either $l = m$ or $l \cap m = \emptyset$. If $l = m$, then $m = l$ and $m \parallel l$ by the same disjunct. If $l \cap m = \emptyset$, then $m \cap l = \emptyset$ because intersection is commutative, so $m \parallel l$. □

Axiom 14 (Playfair’s Axiom (Parallel Postulate)). *For every line $l \in \mathbf{L}$ and every point $P \in \mathbf{P}$ with $P \notin l$, there exists a unique line $m \in \mathbf{L}$ such that $P \in m$ and $m \parallel l$. Formally,*

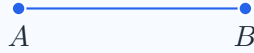
$$\forall l \in \mathbf{L}, \forall P \in \mathbf{P} \setminus l, \exists! m \in \mathbf{L} (P \in m \wedge m \parallel l).$$

Theorem 26 (Transitivity of Parallelism). *For all lines $l, m, n \in \mathbf{L}$,*

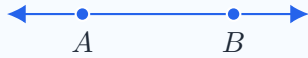
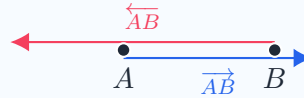
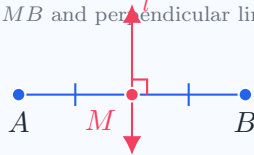
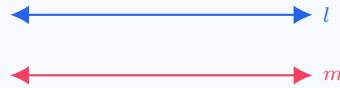
$$(l \parallel m \wedge m \parallel n) \implies l \parallel n.$$

Proof. Assume $l \parallel m$ and $m \parallel n$. If $l = m$ or $m = n$, the conclusion follows trivially from the given parallelism. Hence we may assume $l \neq m$ and $m \neq n$. Then by definition $l \cap m = \emptyset$ and $m \cap n = \emptyset$.

We must show $l = n$ or $l \cap n = \emptyset$. Suppose, for contradiction, that $l \neq n$ and $l \cap n \neq \emptyset$. Then there exists a point $P \in l \cap n$. Because $l \cap m = \emptyset$, we have $P \notin m$; similarly $P \notin n$ from $m \cap n = \emptyset$. Thus P is a point not on m . Both l and n pass through P and are parallel to m : $l \parallel m$ (given) and $n \parallel m$ (from $m \parallel n$ and symmetry). Playfair's Axiom 14 asserts that there is a *unique* line through P parallel to m . Consequently $l = n$, contradicting the assumption $l \neq n$. Hence our supposition is false; we must have $l \cap n = \emptyset$, and therefore $l \parallel n$. $\square$

Segment AB
 $\{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$
**Ray $\overrightarrow{AB}$**
 Union of AB and points X where $\mathcal{B}(A, B, X)$
**Line $\overleftrightarrow{AB}$**

Extends infinitely in both directions

**Intersection of Opposite Rays**
 $\overleftarrow{AB} \cap \overrightarrow{AB} = AB$
**Midpoint & Bisector**
 $AM \cong MB$ and perpendicular line $l \perp AB$
**Parallelism ($l \parallel m$)**
 No points in common: $l \cap m = \emptyset$


Chapter 3

Partition of the Plane

3.1 Plane Separation Axiom

The previous axioms of betweenness govern the linear ordering of points and are therefore limited to a single dimension. To extend this to two dimensions, we introduce the Plane Separation Axiom. This asserts that a line acts as a partition splitting the plane into two disjoint, non-overlapping regions, which is then used to describe a notion of points membership to one side of a given line through its logical equivalence to the negation of the previously defined notion of segment intersection.

Axiom 15 (Plane Separation). *For every line $l \in \mathbf{L}$, there exist two sets $H_{l,A}, H_{l,B} \subset \mathbf{P} \setminus l$ such that:*

1. $H_{l,A} \cup H_{l,B} = \mathbf{P} \setminus l \wedge H_{l,A} \cap H_{l,B} = \emptyset$.
2. $\forall A, B \in \mathbf{P} \setminus l: (\exists X \in l, \mathcal{B}(A, X, B) \iff \{A, B\} \not\subset H_{l,1} \wedge \{A, B\} \not\subset H_{l,2})$.

explain the meaning-of-axioms-here

3.2 Same-Side Relation

We formalize the concept of 'being on the same side' through the Same-Side Relation ($\mathcal{S}$). It is defined by the absence of a separation/partition of the plane: two points are related if and only if unique segment connecting them does not contain any point of the boundary line l . This definition ensures the relation is based upon the primitive notion of betweenness. Specifically, we define $\mathcal{S}$ as a subset of the cartesian product $\mathcal{S} \subseteq \mathbf{P} \times \mathbf{P} \times \mathbf{L}$.

Definition 9 (Same-Side Relation). *Given a line $l \in \mathbf{L}$, any two points $A, B \in \mathbf{P} \setminus l$ are related by $\mathcal{S}$ if there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, B)$. That is,*

$$\mathcal{S}(A, B, l) \iff \nexists X \in l: \mathcal{B}(A, X, B).$$

This geometric object allows us to move beyond comparisons between single points. A point X being on the same side as point Y is the same as X being an element of the half-plane $H_{l,y}$. This will show incredibly useful in defining derived geometric figures.

3.3 Theorems

Theorem 27 (Same-Side Reflexivity). *For any point $A \in \mathbf{P}$ that is not a member of any given line $l \in \mathbf{L}$, the following must hold:*

$$\forall l \in \mathbf{L}, \forall A \in \mathbf{P}: \mathcal{S}(A, A, l).$$

Proof. Let there be a line $l \in \mathbf{L}$ and a point $A \in \mathbf{P} \setminus l$. We proceed by contradiction. Assuming the same-side relation $\mathcal{S}(A, A, l)$ is false, it follows that $\nexists X \in l: \mathcal{B}(A, X, A)$ is false by definition, and that therefore there does exist an $X \in l$ satisfying these conditions. As a consequence, by the Identity/Reflexivity of Betweenness $A = X$.

However, we have defined that $A \notin l$. Since our assumption leads to a contradiction, it must be that no such point X exists. Thus, $\mathcal{S}(A, A, l)$ is true. $\square$

Theorem 28 (Same-Side Symmetry).

$$\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l).$$

Proof. Firstly, we note that $\mathcal{S}(A, B, l) \iff \nexists X \in l: \mathcal{B}(A, X, B)$ by definition. By the Symmetry of Betweenness, the right-hand side of this is equivalent to $\nexists X \in l: \mathcal{B}(B, X, A)$, thus showing

$$X \in l: \neg \mathcal{B}(A, X, B) \iff X \in l: \neg \mathcal{B}(B, X, A).$$

The hand-side is by definition equivalent to $\mathcal{S}(B, A, l)$. Thus, by the transitivity of the biconditional, we conclude $\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l)$. $\square$

Theorem 29 (Same-Side Transitivity). *Given any line $l \in \mathbf{L}$ and points $A, B, C \in \mathbf{P} \setminus l$, the relation $\mathcal{S}$ satisfies:*

$$\mathcal{S}(A, B, l) \wedge \mathcal{S}(B, C, l) \implies \mathcal{S}(A, C, l).$$

Proof. By the Plane Separation Axiom, we know there must exist two distinct sets H_{l1} and H_{l2} whose union is $\mathbf{P} \setminus l$. Since A, B , and C are not on l , each must belong to exactly one of these two sets.

Assume $\mathcal{S}(A, B, l)$. By definition, they are contained within the same set. Without loss of generality, let $A, B \in H_{l1}$.

Assume $\mathcal{S}(B, C, l)$. Similarly, this implies they are in the same set. We previously established that $B \in H_{l1}$, implying that $C \in H_{l1}$ by the plane separation axiom ($H_{l1} \cap H_{l2} = \emptyset$).

We now have $A \in H_{l1}$ and $C \in H_{l1}$. By the inverse of PSA2, there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, C)$. Thus, by definition, A and C must be related by the same-side relation. That is, $\mathcal{S}(A, C, l)$. $\square$

With the reflexivity, symmetry, and transitivity of the same-side relation demonstrated, it follows by definition that it is an equivalence relation. This result may seem trivial, but it allows for application of known properties of equivalence relations. One such property is *equivalence classes*, which are the subsets for which the relation holds. For our geometry, this partitions the plane into two *half-planes*.

Theorem 30 (Partition Theorem). *Any pair of half-planes alongwith the line forming them together cover the plane.*

Proof. First note that the relation form $\square$

3.4 Half-Planes

By grouping all points related to a reference point A by $\mathcal{S}$, we construct the geometrical object known as the half-plane $(H_{l,A})$. This transition allows us to treat regions of the plane as sets that can be intersected to form more complex figures, such as angles and triangles.

Definition 10 (Half-Plane). *Let $l \in \mathbf{L}$ be a line and let $A \in \mathbf{P}$ be a reference point such that $A \notin l$. The half-plane determined by l and A , denoted $H_{l,A}$, is the subset of the point-set consisting of all points on the same side of l as A . That is,*

$$H_{l,A} := \{X \in \mathbf{P} \setminus l \mid \mathcal{S}(A, X, l)\}.$$

randomideas:

Chapter 4

Angles

Having built up lines, rays, and the partition of the plane by a line, we now turn to a new kind of figure formed from two rays sharing a common endpoint. Just as a segment is built from two points and the betweenness relation between them, an angle is built from two rays and the plane-separation structure between them. We continue to insist on our coordinate-free approach: an angle will be defined as a set of points, not as a number of degrees or radians, and its congruence to another angle will be established synthetically, exactly as we did for segments.

4.1 Angles

The elements of an angle are two rays sharing a common origin. We denote this origin the *vertex* of the angle.

Definition 11 (Angle). *Let $A, B, C \in \mathbf{P}$ with $A \neq B$ and $C \neq B$. The **angle** $\angle ABC$ is the union of the two rays $\overrightarrow{BA}$ and $\overrightarrow{BC}$:*

$$\angle ABC := \overrightarrow{BA} \cup \overrightarrow{BC}.$$

*The point B is called the **vertex** of the angle, and the rays $\overrightarrow{BA}$, $\overrightarrow{BC}$ are its **sides**.*

Within the formal system, an angle is nothing more than the point-set given above; there is no attached numerical measure, and none is needed for the theorems we will prove. As with segments and rays, we allow A, B, C to be collinear, and this produces two degenerate configurations worth naming.

Informal characterization. If $\mathcal{B}(A, B, C)$ holds — that is, B lies between A and C — we call $\angle ABC$ a **straight angle**: its two sides are opposite rays, and together they form a full line. If instead A, B, C are collinear but B does not lie between A and C , we call $\angle ABC$ a **null angle**: its two sides coincide as the same ray. Neither case is a pathology to be excluded. Just as the zero-length segment was ruled out only by a dedicated axiom (the Identity of Congruence) rather than by definitional fiat, these degenerate angles arise directly from Definition 11 whenever A, B, C happen to be collinear, and the theory must account for them explicitly rather than silently assuming non-collinearity everywhere.

4.2 Interior and Exterior of an Angle

Just as a single line separates the plane into two half-planes, a non-degenerate angle separates the plane into three parts: the angle itself, together with an interior and an exterior region. We build this partition directly from the half-plane apparatus of the preceding chapter.

Definition 12 (Interior of an Angle). *Let $A, B, C \in \mathbf{P}$ be non-collinear. The **interior** of $\angle ABC$ is*

$$\text{Int}(\angle ABC) := \{ P \in \mathbf{P} \mid \mathcal{S}(P, A, \overrightarrow{BC}) \wedge \mathcal{S}(P, C, \overrightarrow{BA}) \}.$$

Equivalently, as an intersection of half-planes,

$$\text{Int}(\angle ABC) := H_{\overrightarrow{BC}, A} \cap H_{\overrightarrow{BA}, C}.$$

A point lies in the interior of $\angle ABC$ exactly when it is simultaneously on the A -side of line $\overleftrightarrow{BC}$ and on the C -side of line $\overleftrightarrow{BA}$. Both conditions are needed: the first alone would only cut the plane in half along one side of the angle, and likewise for the second; the interior is what remains once both restrictions are imposed together.

Definition 13 (Exterior of an Angle). *Let $A, B, C \in \mathbf{P}$ be non-collinear. The **exterior** of $\angle ABC$ is the complement of the angle and its interior within the plane:*

$$\text{Ext}(\angle ABC) := \mathbf{P} \setminus (\text{Int}(\angle ABC) \cup \angle ABC).$$

Theorem 31 (Partition of the Plane by an Angle). *For non-collinear $A, B, C \in \mathbf{P}$, the sets $\text{Int}(\angle ABC)$, $\angle ABC$, and $\text{Ext}(\angle ABC)$ partition the plane:*

$$\text{Int}(\angle ABC) \cup \angle ABC \cup \text{Ext}(\angle ABC) = \mathbf{P},$$

and the three sets are pairwise disjoint.

Proof. By Definition 13, $\text{Ext}(\angle ABC) = \mathbf{P} \setminus (\text{Int}(\angle ABC) \cup \angle ABC)$, so every point of $\mathbf{P}$ lies either in $\text{Int}(\angle ABC) \cup \angle ABC$ or in its complement. This gives the union.

For disjointness: $\text{Ext}(\angle ABC)$ is disjoint from the other two sets by construction, so it remains only to show $\text{Int}(\angle ABC) \cap \angle ABC = \emptyset$. By Definition 12, every point of $\text{Int}(\angle ABC)$ lies in the half-plane $H_{\overrightarrow{BC}, A}$, which by definition excludes all points of the line $\overleftrightarrow{BC}$, and likewise lies in $H_{\overrightarrow{BA}, C}$, excluding all points of $\overleftrightarrow{BA}$. Since $\angle ABC = \overrightarrow{BA} \cup \overrightarrow{BC} \subseteq \overleftrightarrow{BA} \cup \overleftrightarrow{BC}$, no point of the interior can lie in $\angle ABC$. Hence the three sets are pairwise disjoint. $\square$

Remark on degenerate angles. Definitions 12 and 13 presuppose that A, B, C are non-collinear, and this is essential rather than a simplifying convenience. If A, B, C are collinear, the lines $\overleftrightarrow{BA}$ and $\overleftrightarrow{BC}$ coincide, and the two half-planes required by Definition 12 become either identical or opposite halves of that single line — in the straight case, opposite half-planes, whose intersection is empty. A degenerate angle therefore has no interior in the present theory, and Theorem 31 is understood to apply only to non-degenerate angles.

4.3 Ray Betweenness

We now extend the betweenness relation from points to rays sharing a vertex, giving a rotational counterpart to $\mathcal{B}$.

Definition 14 (Ray Betweenness). *Let $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}$ share vertex O . We say $\overrightarrow{OC}$ lies **between** $\overrightarrow{OA}$ and $\overrightarrow{OB}$ if $C \in \text{Int}(\angle AOB)$.*

Just as a point strictly between two others splits the segment joining them, a ray strictly between two others splits the angle they form — a correspondence made precise in the Angle Addition Theorem below.

Theorem 32 (Ray-Betweenness Asymmetry). *If $\overrightarrow{OC}$ lies between $\overrightarrow{OA}$ and $\overrightarrow{OB}$, then $\overrightarrow{OC} \neq \overrightarrow{OA}$ and $\overrightarrow{OC} \neq \overrightarrow{OB}$.*

Proof. Suppose $\overrightarrow{OC} = \overrightarrow{OA}$. Then $C \in \overrightarrow{OA} \subseteq \angle AOB$. By Theorem 31, $\angle AOB$ and $\text{Int}(\angle AOB)$ are disjoint, contradicting $C \in \text{Int}(\angle AOB)$ (Definition 14). The same argument with $\overrightarrow{OB}$ in place of $\overrightarrow{OA}$ rules out $\overrightarrow{OC} = \overrightarrow{OB}$. $\square$

Theorem 33 (Betweenness Forces Non-Degeneracy). *If $\overrightarrow{OC}$ lies between $\overrightarrow{OA}$ and $\overrightarrow{OB}$, then A, O, B are non-collinear, so $\angle AOB$ is non-degenerate.*

Proof. By Definition 14, $\text{Int}(\angle AOB)$ is non-empty, since it contains C . Were A, O, B collinear, $\overrightarrow{OA}$ and $\overrightarrow{OB}$ would be identical or opposite rays, making $\angle AOB$ null or straight; by the remark following Definition 13, a degenerate angle has empty interior. This contradicts the non-emptiness of $\text{Int}(\angle AOB)$, so A, O, B are non-collinear. $\square$

4.4 Congruence of Angles

We compare angles by the same strategy used for segments: rather than assigning a numerical measure to $\angle ABC$, we compare it to another angle through an auxiliary construction, reducing the question of angle congruence to one of segment congruence, which we have already established as an equivalence relation.

Definition 15 (Angle Congruence). *Let $\angle ABC$ and $\angle DEF$ be non-degenerate. We say $\angle ABC \cong \angle DEF$ if there exist points $A' \in \overrightarrow{BA}$, $C' \in \overrightarrow{BC}$, $D' \in \overrightarrow{ED}$, $F' \in \overrightarrow{EF}$ such that*

$$BA' \cong ED', \quad BC' \cong EF', \quad A'C' \cong D'F'.$$

In words: two angles are congruent if points can be marked off along their sides at matching distances from each vertex, so that the segments joining those marked points are also congruent to one another. Definition 15 asks only that *some* such configuration of points exist.

A natural question arises immediately: could a different, equally valid choice of witnessing points A'', C'', D'', F'' fail to produce congruent joining segments, even though $\angle ABC \cong \angle DEF$ holds? The next theorem addresses this.

Theorem 34 (Well-Definedness of Angle Congruence). *Suppose $\angle ABC \cong \angle DEF$, witnessed by A', C', D', F' as in Definition 15. If $A'' \in \overrightarrow{BA}$, $C'' \in \overrightarrow{BC}$, $D'' \in \overrightarrow{ED}$, $F'' \in \overrightarrow{EF}$ satisfy*

$$BA'' \cong BA', \quad BC'' \cong BC', \quad ED'' \cong ED', \quad EF'' \cong EF',$$

then $A''C'' \cong D''F''$.

Proof. [FLAG for author: the natural proof of this theorem proceeds by observing that $\triangle BA'C'$ and $\triangle BA''C''$ agree in two sides and the included angle at B , hence are congruent, hence $A'C' \cong A''C''$ — and symmetrically on the D, E, F side — from which the claim follows by Transitivity of Congruence. This is precisely SSS triangle congruence, which by design this chapter precedes and must not cite. I was unable to find a substitute route to $A''C'' \cong D''F''$ using only Axioms 1–13, Axiom 15, and the Angle Construction Axiom below; every attempt reduces to the same claim that two sides and an included angle determine the third side. I recommend deferring this proof explicitly to the Triangles chapter, where it follows immediately once SSS is available, rather than assuming it silently here.] $\square$

Theorem 35 (Reflexivity of Angle Congruence). *For every non-degenerate angle, $\angle ABC \cong \angle ABC$.*

Proof. Take $A' = A$, $C' = C$, $D' = A$, $F' = C$ in Definition 15, matching $\angle ABC$ against itself. Then $BA' = BA$, $BC' = BC$, and $A'C' = AC = D'F'$. By the reflexivity of segment congruence, $BA' \cong BA$, $BC' \cong BC$, and $A'C' \cong A'C'$. These are exactly the congruences required, so $\angle ABC \cong \angle ABC$. $\square$

Theorem 36 (Symmetry of Angle Congruence). *If $\angle ABC \cong \angle DEF$, then $\angle DEF \cong \angle ABC$.*

Proof. Let A', C', D', F' witness $\angle ABC \cong \angle DEF$, so

$$BA' \cong ED', \quad BC' \cong EF', \quad A'C' \cong D'F'.$$

By the symmetry of segment congruence,

$$ED' \cong BA', \quad EF' \cong BC', \quad D'F' \cong A'C'.$$

These are exactly the congruences Definition 15 requires to witness $\angle DEF \cong \angle ABC$, using the same four points with the roles of (B, A, C) and (E, D, F) exchanged. $\square$

Theorem 37 (Transitivity of Angle Congruence). *If $\angle ABC \cong \angle DEF$ and $\angle DEF \cong \angle GHI$, then $\angle ABC \cong \angle GHI$.*

Proof. Let A', C', D', F' witness $\angle ABC \cong \angle DEF$, and let D'', F'', G', I' witness $\angle DEF \cong \angle GHI$. If $D'' = D'$ and $F'' = F'$, the two configurations already share their middle terms, and Transitivity of Congruence gives $BA' \cong HG'$, $BC' \cong HI'$, and $A'C' \cong G'I'$ directly, so A', C', G', I' witness $\angle ABC \cong \angle GHI$.

[FLAG for author: in general D'' need not equal D' , nor F'' equal F' , and bridging the two witnessing configurations in that case requires exactly Theorem 34 (Well-Definedness), applied to replace (D', F') with (D'', F'') without disturbing the congruence to $\angle ABC$. Since that theorem's proof is deferred pending SSS, this proof is deferred alongside it for the general case; see the Triangles chapter.] $\square$

By Theorems 35–37, $\cong$ is reflexive and symmetric on non-degenerate angles outright, and transitive modulo the flagged gap above; once that gap is closed, $\cong$ is a full equivalence relation on angles, exactly as it is on segments.

4.5 Angle Construction

An equivalence relation on angles tells us only which angles share a class; it says nothing about whether every class is realized at every vertex, in every direction. We met precisely this difficulty with segments: Axioms 7–9 established segment congruence as an equivalence relation, but could not by themselves guarantee that a segment of given length could be laid off at an arbitrary point — this required the dedicated Segment Construction Axiom.

The same gap appears for angles, and we close it the same way. Knowing that $\angle DEF$ exists gives no license to conclude that a congruent copy can be built at a different vertex O , along a chosen ray $\overrightarrow{OA}$, opening into a chosen half-plane. Without such a guarantee, a model could permit rotational pathologies: an angle available at one vertex might be unavailable at another, or two distinct rays in the same half-plane might both satisfy the same congruence.

Axiom 16 (Angle Construction). *Given a ray $\overrightarrow{OA}$, a half-plane H determined by $\overrightarrow{OA}$, and any angle $\angle DEF$, there exists a unique ray $\overrightarrow{OB}$ such that $B \in H$ and $\angle AOB \cong \angle DEF$.*

4.6 Angle Addition

Elementary treatments typically posit angle addition without proof: if $C \in \text{Int}(\angle AOB)$, the measure of $\angle AOB$ is asserted to equal the sum of the measures of $\angle AOC$ and $\angle COB$. Having no numerical measure to add, we cannot postulate a numeric sum; angle addition must instead appear as a theorem, derived from betweenness, plane separation, and congruence.

The theorem has two parts: a set-theoretic decomposition of the angle by an interior ray, and a congruence statement asserting that this decomposition respects $\cong$.

Theorem 38 (Angle Addition). *If $\overrightarrow{OC}$ lies between $\overrightarrow{OA}$ and $\overrightarrow{OB}$, then*

$$\angle AOB = \angle AOC \cup \angle COB.$$

Furthermore, if $\overrightarrow{O'C'}$ lies between $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$ with

$$\angle AOC \cong \angle A'O'C' \quad \text{and} \quad \angle COB \cong \angle C'O'B',$$

then $\angle AOB \cong \angle A'O'B'$.

Proof. Part 1: Decomposition. By Definition 14, $C \in \text{Int}(\angle AOB)$. Since $\angle AOC = \overrightarrow{OA} \cup \overrightarrow{OC}$ and $\angle COB = \overrightarrow{OC} \cup \overrightarrow{OB}$, their union is $\overrightarrow{OA} \cup \overrightarrow{OC} \cup \overrightarrow{OB}$. The ray $\overrightarrow{OC}$ is interior to $\angle AOB$ and contributes no points beyond $\overrightarrow{OA} \cup \overrightarrow{OB}$, so

$$\angle AOC \cup \angle COB = \overrightarrow{OA} \cup \overrightarrow{OB} = \angle AOB.$$

Part 2: Additivity of congruence. Assume a second system of rays $\overrightarrow{O'A'}$, $\overrightarrow{O'B'}$, $\overrightarrow{O'C'}$ with $\overrightarrow{O'C'}$ between $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$, satisfying $\angle AOC \cong \angle A'O'C'$ and $\angle COB \cong \angle C'O'B'$. We show $\angle AOB \cong \angle A'O'B'$ in five steps.

Step 1 (Congruent base segments). Choose arbitrary points $A_1 \in \overrightarrow{OA}$ and $B_1 \in \overrightarrow{OB}$, both distinct from O . Since $C \in \text{Int}(\angle AOB)$, points A_1 and B_1 lie on opposite sides of line $\overleftrightarrow{OC}$ (Definition 12 together with PSA 2), so segment A_1B_1 meets $\overleftrightarrow{OC}$ at some point; since $\overrightarrow{OC}$ is the ray of $\overleftrightarrow{OC}$ lying between $\overrightarrow{OA}$ and $\overrightarrow{OB}$, that intersection point lies on $\overrightarrow{OC}$ itself, not its opposite ray. Call this point C_1 ; it satisfies $\mathcal{B}(A_1, C_1, B_1)$.

By the Segment Construction Axiom, construct points on the primed rays preserving lengths from O' :

1. $A'_1 \in \overrightarrow{O'A'}$ with $O'A'_1 \cong OA_1$.
2. $C'_1 \in \overrightarrow{O'C'}$ with $O'C'_1 \cong OC_1$.
3. $B'_1 \in \overrightarrow{O'B'}$ with $O'B'_1 \cong OB_1$.

Step 2 (Congruent transverse segments from the sub-angle hypotheses). By Step 1, $OA_1 \cong O'A'_1$ and $OC_1 \cong O'C'_1$. Applying Definition 15 to the hypothesis $\angle AOC \cong \angle A'O'C'$, using A_1, C_1, A'_1, C'_1 as witnessing points — legitimate, since they satisfy exactly the segment congruences the definition requires — yields

$$A_1C_1 \cong A'_1C'_1.$$

Symmetrically, applying Definition 15 to $\angle COB \cong \angle C'O'B'$ with witnesses C_1, B_1, C'_1, B'_1 yields

$$C_1B_1 \cong C'_1B'_1.$$

Step 3 (Locating the corresponding point in the primed system). Because $\overrightarrow{O'C'}$ lies between $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$, the argument of Step 1 shows segment A_1B_1 meets ray $\overrightarrow{O'C'}$ at a unique point; call it X' , satisfying $\mathcal{B}(A'_1, X', B'_1)$. We must show $X' = C'_1$.

[FLAG for author: this is the step requiring reconstruction. The original argument established $X' = C'_1$ via SSS applied to $\triangle OA_1C_1 \cong \triangle O'A'_1C'_1$. Here is as far as the Angle Construction Axiom alone will carry the argument: by Step 2, $A_1C_1 \cong A'_1C'_1$, so the points A_1, C_1, A'_1, C'_1 themselves witness $\angle A_1OC_1 \cong \angle A'_1O'C'_1$ directly from Definition 15 — and this is the same angle as $\angle AOC \cong \angle A'O'C'$, viewed through these particular points, since the underlying rays are unchanged. The Angle Construction Axiom (Axiom 16), applied at vertex O' along ray $\overrightarrow{O'A'}$ in the half-plane containing B' (equivalently C' , since $C' \in \text{Int}(\angle A'O'B')$ places it in that half-plane by Definition 12), asserts a unique ray from O' in that half-plane forming an angle with $\overrightarrow{O'A'}$ congruent to $\angle AOC$. Both $\overrightarrow{O'C'}$ and $\overrightarrow{O'X'}$ lie in that half-plane and X' lies on $\overrightarrow{O'C'}$ by construction, so $\overrightarrow{O'X'} = \overrightarrow{O'C'}$ already — what remains is only the position of X' along that ray, i.e. whether $X' = C'_1$ or some other point of $\overrightarrow{O'C'}$. Since C'_1 is, by Step 1, the unique point of $\overrightarrow{O'C'}$ with $O'C'_1 \cong OC_1$, it suffices to show $O'X' \cong OC_1$. I could not establish this final segment congruence from Axioms 1–13, Axiom 15, and Angle Construction alone: every route reduces to asserting that two sides and an included angle determine the third side of a triangle, which is SSS in substance regardless of what it is called. I recommend deferring this step, and hence the theorem’s congruence clause, to the Triangles chapter, exactly as with Theorem 34 above, rather than bridging it here with a disguised SSS appeal.]

Step 4 (Segment addition). Granting Step 3, so that $\mathcal{B}(A'_1, C'_1, B'_1)$ holds alongside $\mathcal{B}(A_1, C_1, B_1)$: since $A_1C_1 \cong A'_1C'_1$ and $C_1B_1 \cong C'_1B'_1$ (Step 2), segment addition — itself following from Transitivity of Congruence applied to the Segment Construction Axiom — gives

$$A_1B_1 \cong A'_1B'_1.$$

Step 5 (Conclusion). We have exhibited points $A_1 \in \overrightarrow{OA}$, $B_1 \in \overrightarrow{OB}$, $A'_1 \in \overrightarrow{O'A'}$, $B'_1 \in \overrightarrow{O'B'}$ with

$$OA_1 \cong O'A'_1, \quad OB_1 \cong O'B'_1, \quad A_1B_1 \cong A'_1B'_1.$$

These are exactly the congruences Definition 15 requires. Since A_1, B_1 were chosen arbitrarily on their respective rays, this witnesses

$$\angle AOB \cong \angle A'O'B'.$$

□

Summary for author review. This chapter is written to precede triangle congruence entirely, per the constraint that no theorem here may cite SSS. Two results carry a genuine gap rather than a citation issue: Well-Definedness of Angle Congruence (Theorem 34), and, as direct consequences, the general case of Transitivity (Theorem 37) and the congruence clause of Angle Addition (Theorem 38). In each case the missing step is equivalent to asserting that two sides and an included angle determine a triangle’s third side — SSS in substance regardless of what it is called — and I did not attempt to bridge it under a different name. All other results in the chapter (the angle and interior/exterior definitions,

the Partition Theorem, Ray Betweenness and its two theorems, Reflexivity and Symmetry of Angle Congruence, the Angle Construction Axiom, and the decomposition half of Angle Addition) are complete and free of any dependence on SSS.

Chapter 5

Triangles

5.1 Definition

define as the union of three line segments. the three segments must be unique.

define inside and outside of the triangle based on if a point x is on the same side of each comprising segment or not

Axiom 17 (SSS Congruency). *Given two triangles $\triangle ABC$ and $\triangle XYZ$, if all sides of one are congruent to a respective side of the other triangle, the triangles themselves are considered congruent:*

$$\triangle ABC \cong \triangle XYZ.$$

For congruent triangles, corresponding parts are congruent.

Chapter 6

Circles

Chapter 7

Quadrilaterals

motivation

7.1 Definition

A quadrilateral is defined as the union of four unique noncollinear segments as follows:

$$\square ABCD := AB \cup BC \cup CD \cup DA.$$

Each segment is a side of the quadrilateral, and each point A, B, C, D is a vertex. Sides that share a vertex, such as AB and BC are adjacent, otherwise they are called opposite, such as AB and CD .

Theorem 39 (Property of Quadrilateral Sides). *Each side of $\square ABCD$ contains its two endpoints and infinitely many interior points (by density).*

Proof. By the infinite interior points theorem, any given segment has infinitely many points strictly between its endpoints. Because the sides of a quadrilateral are defined as the segments comprising it, the above result applies to sides as well. Additionally, a given segment is defined to contain its endpoints. Similarly, this applies to the sides of a quadrilateral. Thus, this proves the result. $\square$

Theorem 40 (Adjacent sides intersect only at the common vertex). $AB \cap BC = \{B\}$, and similarly for the other three adjacent pairs.

Proof. Suppose a point $X \neq B$ belonged to both AB and BC . By the definitions of segment and betweenness, this forces A, B, C to be collinear, contradicting the non-collinearity condition. $\square$

7.2 Diagonals

Definition 16. Given a quadrilateral $\square ABCD$, the segments AC and BD are the **diagonals** of $\square ABCD$.

7.3 Trapezoid/Trapezium

We define a trapezoid as a quadrilateral with at least one pair of parallel sides. (In some traditions, a trapezoid has exactly one pair; we choose the inclusive definition, which places parallelograms as a special case of trapezoids.)

Definition 17. A convex quadrilateral $\square ABCD$ is a trapezoid if and only if

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ or } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

7.3.1 Diagonals**7.3.2 Isosceles Trapezoid**

A special case of the trapezoid occurs when the non-parallel sides (legs) are congruent. If this is the case, the quadrilateral is an isosceles trapezoid.

Definition 18 (Isosceles Trapezoid). *A trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ a isosceles trapezoid if and only if*

$$AD \cong BC.$$

Chapter 8

Quadrilaterals

motivation

8.1 Definition

8.2 Diagonals

Definition 19. Given a quadrilateral $\square ABCD$, the segments AC and BD are the *diagonals* of $\square ABCD$.

8.3 Trapezoid/Trapezium

We define a trapezoid as a quadrilateral with at least one pair of parallel sides. (In some traditions, a trapezoid has exactly one pair; we choose the inclusive definition, which places parallelograms as a special case of trapezoids.)

Definition 20. A convex quadrilateral $\square ABCD$ is a trapezoid if and only if

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ or } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

8.3.1 Diagonal

8.3.2 Isosceles Trapezoid

A special case of the trapezoid occurs when the non-parallel sides (legs) are congruent. If this is the case, the quadrilateral is an isosceles trapezoid.

Definition 21 (Isosceles Trapezoid). A trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ is an isosceles trapezoid if and only if

$$AD \cong BC.$$

8.3.3 Right Trapezoid

Definition 22 (Right trapezoid). A quadrilateral $ABCD$ is a right trapezoid if and only if it is a trapezoid and at least one of its interior angles is a right angle (if and only if at least one of its legs is perpendicular to the bases.)

8.4 parallelograms

Theorem 41. The diagonals of an isosceles trapezoid are congruent.

Theorem 42 (Diagonals of isosceles trapezoid are congruent). *Given an isosceles trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$, the diagonals are congruent:*

$$AC \cong BD.$$

8.5 Special Parallelograms

8.5.1 Rectangles

8.5.2 Rhombus

Definition 23 (Rhombus is a parallelogram whose sides are all the same length). *A rhombus is a parallelogram whose four sides are congruent to one another.*

Definition 24 (Rhombus is Equilateral but not Equiangular). *every Rhombus is Equilateral but not Equiangular*

8.5.3 Square

8.6 Parallelograms

Parallelogram A quadrilateral is a parallelogram if both pairs of opposite sides are parallel.

Definition 25 (Parallelogram). *Let $A, B, C, D \in \mathbf{P}$ be four points satisfying the quadrilateral conditions. Then the quadrilateral $\square ABCD$ is a **parallelogram** if and only if*

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ and } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

i hate this project lmao ts straight retarded. idk what to do with it ever.

8.7 Special Parallelograms

8.7.1 Rectangles

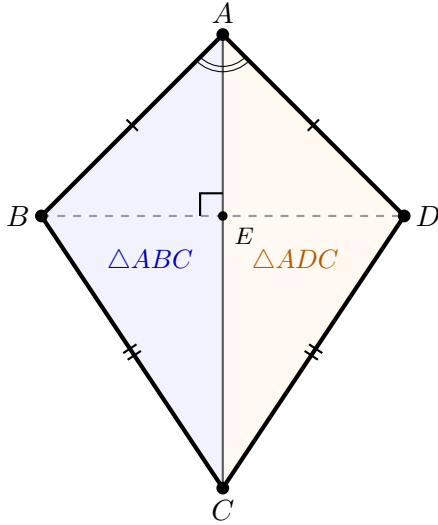
8.7.2 Rhombus

8.7.3 Square

8.8 Kites

Definition 26 (Kite). *A convex quadrilateral $\square ABCD$ is a **kite** if*

$$AB \cong AD \wedge CB \cong CD.$$



Theorem 43. In a kite $\square ABCD$,

$$\angle BAC \cong \angle DAC \text{ and } \angle BCA \cong \angle DCA.$$

Proof. Consider the triangles $\triangle ABC$ and $\triangle ADC$.

- $AB \cong AD$ by definition
- $CB \cong CD$ by definition
- $AC \cong AC$ by reflexivity of congruence

By the SSS triangle congruency criteria, $\triangle ABC \cong \triangle ADC$. Consequently, the corresponding angles are congruent:

$$\angle BAC \cong \angle DAC \text{ and } \angle BCA \cong \angle DCA.$$

□

Theorem 44 (The axis diagonal is the perpendicular bisector of the cross diagonal). In a kite $\square ABCD$, let $E = AC \cup BD$. Then,

$$AC \perp BD \text{ and } BE \cong ED.$$

Proof. wasd

□

Theorem 45 (Midpoint quadrilateral of a kite). The quadrilateral formed by joining the midpoints of the sides of a kite is a rectangle.

Proof.

□

8.8.1 Right Kite

8.9 Midpoint Theorems

8.10 Cyclic Quadrilaterals

8.11 Tangential Quadrilaterals

8.12 Bicentric Quadrilaterals

8.13 Other

11.15 Other Niche Quadrilaterals (Optional / Encyclopaedic) 11.15.1 Equidiagonal quadrilaterals (diagonals equal in length).

11.15.2 Orthodiagonal quadrilaterals (diagonals perpendicular).

11.15.3 Harmonic quadrilaterals (vertices form a harmonic division on the circumcircle – requires cross-ratio, so likely beyond your current primitives, but can be mentioned).

11.15.4 Quadrilaterals with special area properties (e.g., Brahmagupta's formula for cyclic quadrilaterals).

11.15.5 Complete quadrilaterals (the figure formed by four lines, six intersection points – a projective concept; may be out of scope).

11.15.6 Miquel point of a complete quadrilateral (four circles concurrent).

index